

FEDERAL AVIATION ADMINISTRATION

AIR TRAFFIC CONTROL ACADEMY

Statement of Objectives- DRAFT

Air Traffic Performance & Excellence System (APEX)

Purpose

This Statement of Objectives (SOO) describes the Federal Aviation Administration's (FAA) goals and expectations for the implementation and operation of the ATC Training Performance Intelligence System. The FAA seeks concepts, approaches, and general capabilities only; please do not submit proprietary, trade secret, or confidential information. All responses should be written at a level that can be released publicly without restriction. The FAA seeks a scalable, data-driven system capable of providing near-real-time visibility into student performance across all phases of Academy training.

Background

The Federal Aviation Administration (FAA) operates a complex air traffic control system that relies on highly trained controllers to safely and efficiently manage aircraft movements.

The current process for training ATCs follows two tracks. Track One, which is the focus of this SOO, is for new students to attend the FAA Academy in Oklahoma City, OK, for Air Traffic Basics and Initial Qualification training. Track Two is for students who have graduated from a CTI or e-CTI school and do not need to attend the Academy, and are sent directly to an FAA facility.

The ATC Academy is responsible for training Air Traffic Controllers to meet national staffing requirements. The training environment includes:

- Classroom instruction
- Written testing and quizzes
- High-fidelity simulation events
- Instructor evaluations
- Remediation processes
- Phase progression milestones

Objectives

The Federal Aviation Administration requires enhanced visibility into training performance in order to:

- Identify objective-level learning deficiencies
- Identify early risk indicators to support data backed remediation strategies
- Improve retention and mastery
- Ensure standardization of instructional delivery and student assessment standards
- Reduce controller “washout” rates
- Sustain operational safety standards while increasing Academy training throughput
- Ensure performance analytics strengthen safety assurance while supporting national staffing requirements. Inform adjustments to the curriculum using empirical data

Scope of Capability Sought

The FAA seeks a modern, adaptive solution that can serve as an enterprise training performance intelligence backbone. The FAA intends for this solution to function as an enterprise-level training performance intelligence infrastructure, forming a scalable data architecture that integrates disparate training data sources into a unified analytics environment. The system must serve as the foundational performance intelligence backbone for Academy training operations and future enterprise expansion. The FAA does not seek a standalone reporting overlay or content delivery enhancement tool; rather, the solution must establish an interoperable, extensible enterprise data framework capable of supporting predictive modeling, performance standardization, and strategic decision-making. The FAA requests that respondents provide concept papers identifying real-world solutions, along with an implementation plan that, at a minimum, addresses the following areas:

NOTE: The FAA does not request proprietary algorithms, detailed source code, or other confidential technical data. Responses should describe approaches and capabilities at a conceptual level sufficient for evaluation but not include material you would not want publicly disclosed.

A. Comprehensive Data Integration

Demonstrate the solution’s ability to ingest, normalize, and integrate data from:

- Learning Management Systems (LMS)
- Written exams and quizzes
- High-fidelity simulator grading systems
- Simulation scenario event logs
- Instructor scoring rubrics
- Remediation and progression data

- Solutions must minimize manual data entry and support automated data feeds where possible.

B. Objective-Level Mastery Mapping

Demonstrate the solution's ability to:

- Map lessons, blocks, and scenarios to defined objectives
- Associate test questions and simulator events to specific learning objectives
- Track mastery progression at individual and cohort levels
- Provide objective-level proficiency heat maps
- Analyze learner behavior and performance in real-time

C. Near-Real-Time Analytics Dashboards

Demonstrate the solution's ability to provide:

- Individualized student metrics
- Cohort-level analytics
- Instructor-level analytics
- Block and phase summaries
- Executive-level rollups

The FAA requires that these dashboards support drill-down capability from enterprise view to objective-level data.

D. Error Taxonomy and Trend Analytics

Demonstrate the solution's ability to:

- Categorize performance errors by defined taxonomy
- Track frequency and clustering of errors
- Identify trends across cohorts
- Support root-cause analysis of performance degradation

Where applicable, analyze learner interaction data that demonstrably correlates to objective-level performance outcomes. Behavioral analytics shall support training performance improvement and predictive modeling, not serve solely as engagement metrics.

E. Predictive Analytics

Demonstrate the solution's ability to:

- Identify early indicators of student attrition risk and performance degradation=
- Support proactive intervention strategies designed to improve retention and pipeline throughput Forecast cohort-level performance trends
- Correlate early performance metrics to downstream outcomes
- Provide explainable predictive models

F. Instructor Performance Analysis

Demonstrate the solution's ability to:

- Analyze grading patterns by instructor
- Identify scoring variance
- Support instructor calibration efforts to ensure grading criteria are applied consistently, feedback is uniform, and performance standards are interpreted the same way
- Provide remediation frequency analytics

G. Enterprise Leadership Reporting

Demonstrate the solution's ability to provide leadership visibility into:

- Throughput metrics
- Attrition and failure rates
- Time-to-proficiency
- Block-level performance risk
- Cohort comparisons
- Longitudinal trend analysis

H. Technical and Security Requirements

As a reminder: Any description of technical capabilities should be provided in non-proprietary terms. Do not submit sensitive intellectual property, proprietary solution diagrams, or detailed configurations. This RFI seeks general descriptions suitable for public release.

Respondents shall describe:

- Deployment model (cloud/on-premise/hybrid)
- FedRAMP or federal compliance experience

- Authority to Operate (ATO) support capability
- Data ownership structure (FAA retains ownership of all data and models)
- Data export capability in non-proprietary formats
- Approach to avoiding vendor lock-in
- The solution must adhere to FAA requirements for network connectivity, including security, compatibility, and performance requirements

I. Scalability and Future Expansion

Solutions must be scalable to:

- Varying volumes of trainees at the Academy, supporting higher throughput
- Potential expansion to field facilities
- Evolving curriculum requirements
- Future NAS modernization initiatives

J. Implementation Approach

Respondents shall provide:

- High-level architecture diagram
- Data model overview
- Implementation timeline (phased if applicable)
- Approach to minimizing operational disruption
- Estimated implementation cost range
- Ongoing licensing or sustainment cost structure

Nature of this RFI:

(a) This is not a commitment to issue a further screening information request or request for proposals. (b) The FAA is not seeking or accepting unsolicited proposals. (c) The FAA will not pay for any information received or costs incurred in preparing any response to this RFI. (d) Any costs associated with any response to this RFI are solely at the interested vendor's expense. (e) The FAA does not seek and will not review proprietary, trade-secret, or confidential information in response to this RFI. Responses should be written so they can be made publicly available.

Optional:

- Demonstration materials

- Sample dashboards using representative data

Point of Contact: To contact the FAA relating to this RFI, please email the Contracting Officer, Stephen Shaffer, at stephen.c.shaffer@faa.gov. Please, no phone calls.

Submittal Requirements:

All responses must be limited to 25 pages (PDF File, Times New Roman, 12 Point Font) and should include the following: 1. Description of solution and how it will meet the objectives listed in the scope of capability sought 2. Plan for implementation at the FAA Academy. This section must address any proposed customizations to existing products and technology to enable it to satisfy the requirements related to Air Traffic training 3. List any relevant experience applying the proposed solution to training in technically complex fields 4. High-level cost estimate for the proposed solution to be used for research purposes only (non-proprietary and suitable for public release, avoid including proprietary breakdowns or confidential pricing details) 5. Proposed project schedule 6. Contact information responses must be submitted to the Government at the email address above no later than 4 p.m. Eastern time on Monday, April 20th, 2026.

The FAA prefers that no protected or proprietary information be submitted; however, if such data is provided despite this request, respondents must clearly mark the title page with the protective legend cited below. If the response includes data in the response that the respondent does not want to be disclosed to the public or for the FAA to use except for review purposes, please mark the title page with the following legend in compliance with AMS 3.2.2.3-16 “Restricting, Disclosing and Using Data” (April 2024): “This response includes data that must not be (1) disclosed outside the FAA and (2) duplicated, used, or disclosed -in whole or in part- for any purpose other than to review this response.